Advanced Coordinate Geometry

Section 1: Ellipses & Focal Properties

1 What are the coordinates of the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$?

a. $(\pm 4, 0)$

b. $(0, \pm 4)$

c. $(\pm 5, 0)$

d. $(0, \pm 3)$

2 What is the length of the minor axis of the ellipse $\frac{(x-2)^2}{16} + \frac{(y+3)^2}{36} = 1$?

a. 4

b. 8

c. 6

d. 12

3 What is the eccentricity of the ellipse $9x^2 + 25y^2 = 225$?

a. $\frac{4}{5}$

b. $\frac{3}{5}$

c. $\frac{16}{25}$

d. $\frac{5}{4}$

Section 2: Hyperbolas & Asymptotes

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4 What are the equations of the asymptotes of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$?

a. $y = \pm\frac{3}{4}x$

b. $y = \pm\frac{4}{3}x$

c. $y = \pm\frac{9}{16}x$

d. $y = \pm\frac{16}{9}x$

5 What is the distance between the two vertices of the hyperbola $\frac{y^2}{64} - \frac{x^2}{36} = 1$?

a. 12

b. 16

c. 20

d. 24

6 What is the focal length (distance from center to a focus) of $\frac{x^2}{144} - \frac{y^2}{25} = 1$?

a. 11

b. 12

c. 13

d. 17

Section 3: Conic Classification & Discriminants

7 Classify the conic section represented by $3x^2 + 3y^2 - 12x + 6y - 15 = 0$.

a. Circle

b. Ellipse

c. Parabola

d. Hyperbola

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8 Which conic section is represented by the equation $y^2 - 4x - 6y + 1 = 0$?

a. Circle

b. Ellipse

c. Parabola

d. Hyperbola

9 What is the discriminant $B^2 - 4AC$ of the general conic $4x^2 - 12xy + 9y^2 + 2x - 5y = 0$, and what does it indicate?

a. 0 (Parabola)

b. -48 (Ellipse)

c. 48 (Hyperbola)

d. 0 (Degenerate Circle)

Section 4: Rotations & Coordinate Shifts

10 Under a 90° counterclockwise rotation about the origin, what are the new coordinates of the point $(3, -5)$?

a. $(5, 3)$

b. $(-5, -3)$

c. $(3, 5)$

d. $(-3, 5)$

11 A line $y = 3x - 2$ is reflected across the y-axis. What is the equation of the reflected line?

a. $y = -3x - 2$

b. $y = -3x + 2$

c. $y = 3x + 2$

d. $y = -\frac{1}{3}x - 2$

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12 What is the distance from the point $(2, 3)$ to the line $3x - 4y + 1 = 0$?

a. 1

b. 2

c. $\frac{7}{5}$

d. $\frac{5}{7}$

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